

LC 220

Installation and operating instructions



Original installation and operating instructions.

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**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

**Warning**

The use of this product requires experience with and knowledge of the product.

Persons with reduced physical, sensory or mental capabilities must not use this product, unless they are under supervision or have been instructed in the use of the product by a person responsible for their safety.

Children must not use or play with this product.

As the LC 220 controller is part of either a Multilift or Unolift system, there is no separate EC declaration of conformity for LC 220.

Note

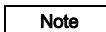
Please see declaration of conformity in the installation and operating instructions for the lifting station.

1. Symbols used in this document**Warning**

If these safety instructions are not observed, it may result in personal injury.



If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Notes or instructions that make the job easier and ensure safe operation.

**Warning**

These instructions must be observed for explosion-proof pumps. It is advisable also to follow these instructions for standard pumps.

2. Scope of delivery

Grundfos LC 220 controllers can be ordered together with wastewater lifting stations, such as Multilift MSS or Unolift. The controller is delivered with mains cable and appropriate plug. An accessories bag containing the following items is also included:

- 1 x installation and operating instructions.

3. Transportation and storage

For long periods of storage, the LC 220 controller must be protected against moisture and heat.

4. Product description

Grundfos LC 220 controllers are designed for control of wastewater lifting stations, such as Multilift MSS or Unolift. The level controller switches the pump on and off according to the liquid level measured by the piezoresistive level sensor. When the start level is reached, the pump will start, and when the liquid level has been lowered to the stop level, the pump will be stopped by the controller. An alarm is indicated in case of high water level in the tank, sensor failure etc.



Fig. 1 LC 220 level controller

The LC 220 controller features the following functions:

- on/off control of the pump based on a continuous signal from the piezoresistive level sensor
- automatic test runs for two seconds (24 hours after last operation) to maintain the pump and its shaft seal
- battery backup for the buzzer in case of mains supply failure (Battery and battery connector available as accessory)
- selection of automatic alarm resetting (via DIP switch)
- selection between two inlet levels (via DIP switch)
- operating indication:
 - power on
 - pump running
 - service/maintenance indication.
- alarm indication:
 - liquid level too high (high-level alarm)
 - wrong phase sequence detected or phase missing (for three-phase pumps only)
 - sensor failure
 - external level alarm
 - runtime monitoring.

The LC 220 controller has one alarm signal output for common alarm and two signal inputs. One input is for connecting an additional float switch working in parallel to the piezoresistive level sensor. The other input is a signal input to trigger optical and acoustical alarms of the controller which could be for instance flood detection outside the lifting station as lifting stations are often installed in a sump inside the basement - the lowest point in the building. In case of e.g. groundwater inflow or water pipe burst, an alarm will be indicated by the controller. If the liquid level in the sump rises above the high-level alarm level, the corresponding alarm LED is on and the built-in buzzer sounds if an external float switch indicates such a situation to the controller. A battery (accessory) can be installed to give an acoustic alarm in case of a power failure.

In case of power failure, the common alarm output, which is a potential-free changeover contact, can forward the alarm signal to a control room by means of an external power source.

Type key, LC 220 controller

Example	LC 220	.1	.230	.1	.8
LC 220 = controller type					
1 = one-pump controller					
Voltage [V]					
1 = single-phase					
3 = three-phase					
Maximum operating current per pump [A]					

Nameplate, LC 220 controller

The actual controller type, voltage variant, etc., are stated in the type designation on the nameplate situated on the side of the controller cabinet.

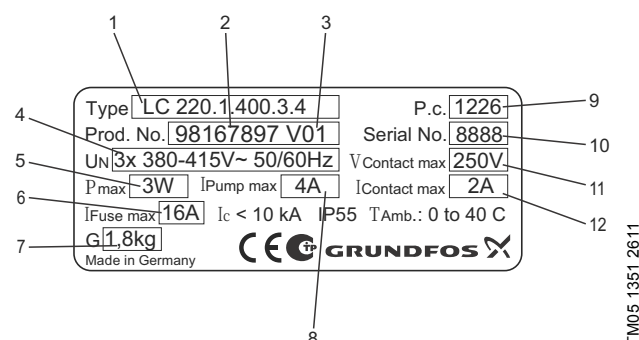


Fig. 2 Example of an LC 220 nameplate

Pos.	Description
1	Type designation
2	Product number
3	Version number
4	Rated voltage
5	Power consumption
6	Maximum backup fuse
7	Weight
8	Maximum pump input current
9	Production year and week
10	Serial number
11	Maximum voltage at contactor
12	Maximum current at contactor

4.1 Design

The LC 220 level controller incorporates the necessary components such as relays and a control panel with indicator lights for indication of operating conditions and fault indications. Furthermore, it has a level input which is activated directly via the pressure tube inside the collecting tank. Finally, it has terminals for power supply, connection to the pump and an alarm signal output for common alarm.

The front cover is closed by four bayonet fastenings with quarter turn locks. On the left side, the locks are extended and connected to the cabinet bottom with hinge strings.

4.2 Control elements and indicator lights

The table below gives a description of the function of the various control elements and indicator lights:

Element	Function	Description
	Selection of operating mode	The operating mode is selected by the ON-OFF-AUTO selector switch which has three different positions: POS I: Starts the pump manually. POS O: - Stops the pump manually - Resets alarm indications. POS AUTO: Automatic operation. The pump will start and stop according to the signal from the level sensor.
	Indication of power supply status	Green indicator light, indicating that the power supply is on.
	Indication of pump status	Red and green indicator lights, indicating pump status: Green: Pump is running. Red: Pump fault.
	High-level alarm	Red indicator light, indicating high water level. The LED lights up if the level sensor measures a certain level in the collecting tank.
	Phase-sequence fault	Red indicator light, indicating wrong phase sequence (three-phase pumps). Change phase sequence by following the instruction in fig. 3.
	Sensor failure alarm	Red indicator light, indicating faulty or missing connection to the level sensor.
	External level alarm	Red indicator light, indicating an alarm from an external level switch.
	Indication of time for service	Yellow indicator light, indicating that it is time for service. This function can be switched on and off by the DIP switch. The factory setting is one year according to EN 12056-4.



Fig. 3 Changing phases of a three-phase controller with phase inverter

4.3 Level sensor

The piezoresistive pressure sensor placed in the controller is connected via a hose to a pressure tube in the tank. The pressure tube extends down into the tank. The rising liquid level compresses the air inside the pressure tube and hose, and the piezoresistive sensor transforms the changing pressure into an analogue signal. The controller uses the analogue signal to start and stop the pump and to indicate high water-level alarm. The pressure tube is fixed underneath a cap and can be taken out for maintenance, service and for cleaning the inside of the tube. Please note that different pressure tubes are used for Multilift and Unolift. Multilifts have a DN 100 tube with a screw cap, whereas Unolift comes with a DN 50 tube with a cap to be inserted.

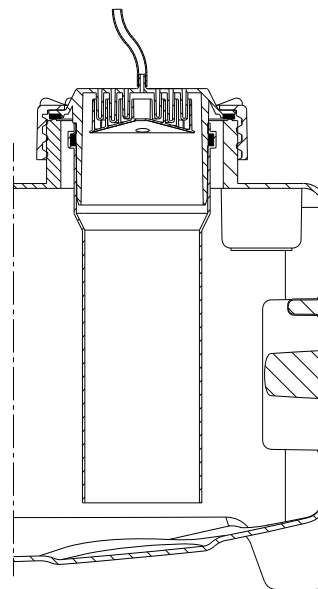


Fig. 4 Pressure tube with hose

5. Installation

Warning
Before making any connections in LC 220 or working on pump, pit, etc., make sure that the power supply has been switched off and that it cannot be accidentally switched on.

The installation must be carried out by authorised personnel in accordance with local regulations.

5.1 Location

Warning
Do not install the LC 220 controller in explosion hazard areas.

LC 220 can be installed at ambient temperatures ranging from 0 °C to +40 °C.

Enclosure class: IP56.

Install the controller as close as possible to the lifting station.

When installed outdoors, LC 220 must be placed in a protective shed or enclosure. LC 220 must not be exposed to direct sunlight.

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5.2 Mechanical installation



Warning

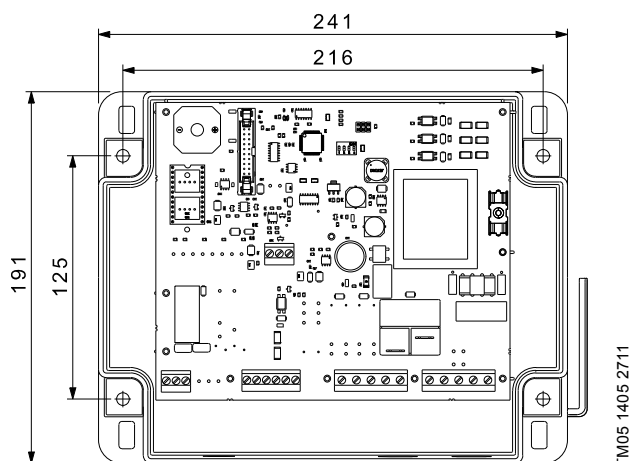
When drilling the holes, take care not to damage any cables or water and gas pipes. Ensure a safe installation.

Note

LC 220 can be mounted without removing the front cover.

Proceed as follows:

- Mount LC 220 on a plane wall surface.
- Mount LC 220 with the cable entries pointing downwards (additional cable entries, if required, must be fitted in the bottom plate of the cabinet).
- Mount LC 220 with four screws through the mounting holes in the back plate of the cabinet. Drill the mounting holes with a 6 mm drill using the drilling template supplied with the controller. Fit the screws into the mounting holes and tighten securely. Fit the plastic caps.



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Fig. 5 Mounting of controller on the wall

5.3 Electrical connection



Warning

LC 220 must be connected in accordance with the rules and standards in force for the application in question.



Warning

Before opening the cover, switch off the mains supply.

The operating voltage and frequency are marked on the controller nameplate. Make sure that the controller is suitable for the electricity supply on which it will be used.

All cables/wires must be fitted through the cable entries and gaskets.

The power supply socket must be placed near the cabinet as the controller is supplied with a 1.5 m cable, a Schuko plug for single-phase or a CEE plug for three-phase pumps.

Maximum backup fuse is stated on the controller nameplate.

If required according to local regulations, install an external mains switch.

5.3.1 Internal layout of LC 220

Figure 6 shows the connectors and the internal layout of LC 220.

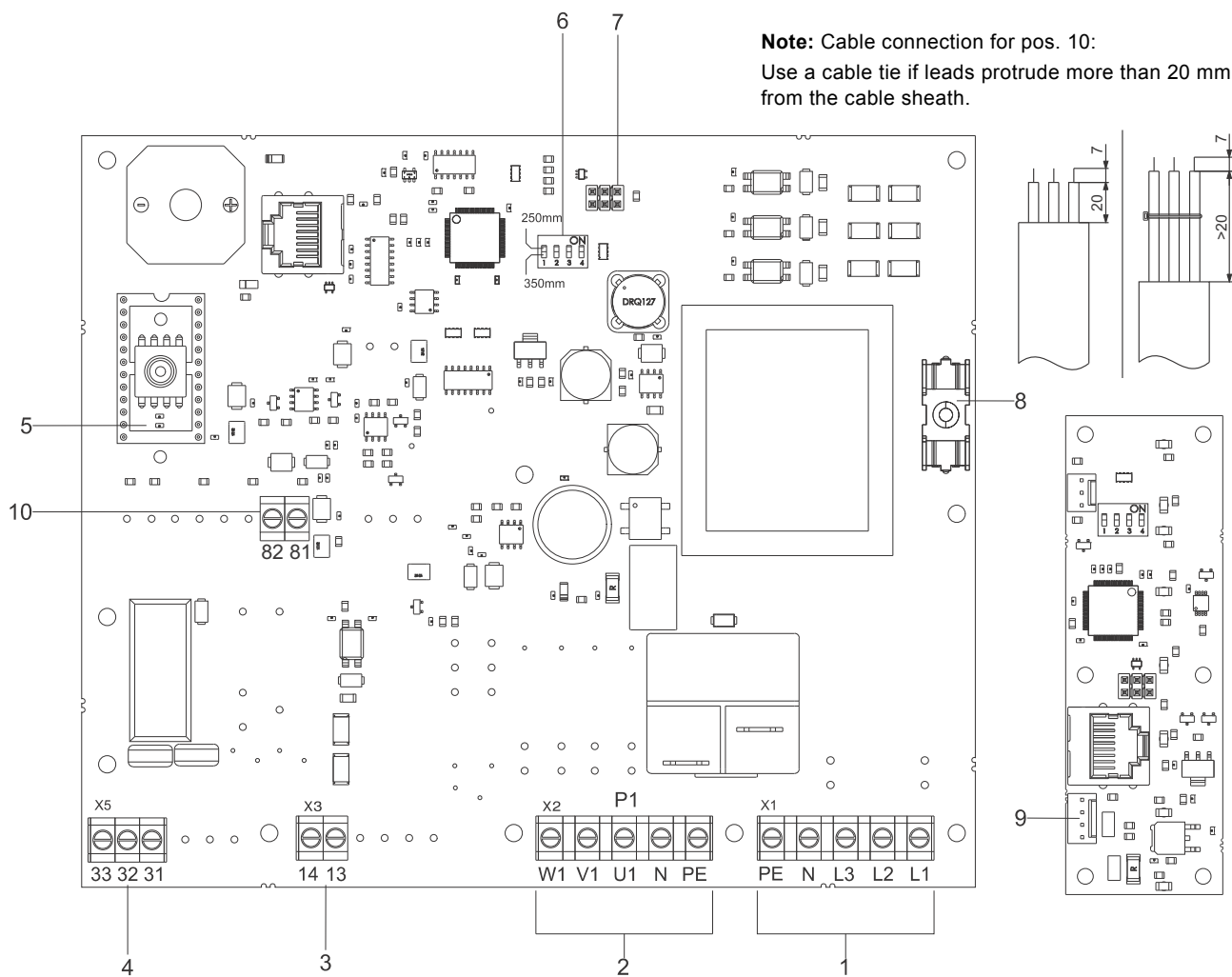


Fig. 6 Internal layout of LC 220

Pos.	Description	Terminal numbers
1	Terminals for power supply	PE, N, L3, L2, L1
2	Terminals for connecting the pump	W1, V1, U1, N, PE
3	Terminals for connecting an external level switch to start the pump.	230 V, NO 13, 14
4	Terminals for "common alarm" output signal	Potential-free changeover contacts NO/NC with max. 250 V / 2 A 31, 32, 33
5	PCB with piezoresistive pressure sensor	0-5 V
6	DIP switch	- Inlet height: ON = 250/520 mm (Multilift MSS/Unolift) OFF = 180 mm - Reset setting: ON = Automatic OFF = Manual - Service interval: ON = 1 year OFF = None - Calibrate level sensor (Do not touch if you do not know how to calibrate) ON = Safe (normal position) OFF = Switch briefly to OFF for sensor to adjust to ambient pressure. (See service instructions)
7	Service connector (PC Tool)	6-pole connector
8	Control circuit fuse, fine-wire fuse	100 mA / 20 mm x Ø5
9	Battery (non-rechargeable)	9 V
10	Terminals for internal float switch	NO 81, 82

NO = Normally Open

NC = Normally Closed

5.4 Connecting the level sensor

The level measurement is done by a pressure sensor located in the controller. The pressure is transmitted to the sensor by means of a transparent pressure hose connected to the pressure tube in the tank.

The pressure hose must be routed rising towards the controller to avoid condensation in the hose.

Connect the pressure hose between the pressure tube in the tank and the bulk head fitting in the controller cabinet. The pressure hose must be inserted into the cabinet up to the stop. Insert approximately 15 mm. Otherwise there is a risk of leakage resulting in pressure loss, inaccurate level detection and malfunction of the system.

Excessive lengths of pressure hose should be cut away.

The pressure hose must not be kinked.

The connection must be air tight to ensure reliable operation.

5.5 Setting

LC 220 has a 4-contact DIP switch. See fig. 7.

To make the settings, open the cabinet cover by means of the bayonet fastener locks. The locks on the left side have hinge strings.

Note

Before starting setting, the controller must be off circuit for at least 10 seconds to ensure the correct configuration during startup after change of the DIP-switch setting.

Set the switches 1 to 4 as follows:

- **Switch 1** (start level):

Pos.	Description
ON	The pump starts when the level in the collecting tank reaches 250 mm (Multilift) or 520 mm (Unolift).
OFF	The pump starts when the level in the collecting tank reaches 180 mm.

- **Switch 2** (automatic alarm reset):

Pos.	Description
ON	The fault indication will be reset after the fault has disappeared, meaning the indicator lights will be turned to off and the alarm signals to external alarm devices and the built-in buzzer will be deactivated.
OFF	The alarm signal must be reset manually by pushing the selector switch to position "0".

- **Switch 3** (service interval):

Pos.	Description
ON	The function for reminding that it is time for maintenance is activated. The yellow indication light will be on when it is time for maintenance work. The maintenance interval is 1 year (fixed value).
OFF	The function is deactivated.

- **Switch 4** (sensor calibration):

Pos.	Description
ON	Must always be in this position (preset).
OFF	For service only.

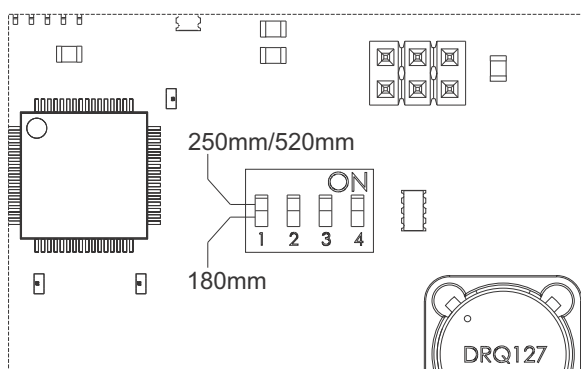


Fig. 7 DIP switch

5.6 Wiring diagrams

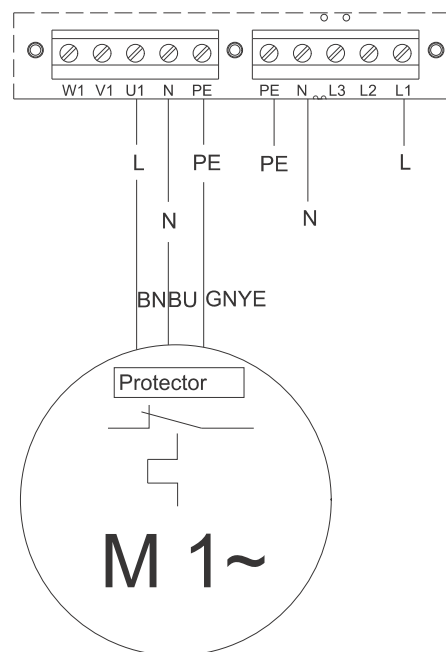


Fig. 8 Wiring diagram for single-phase pump

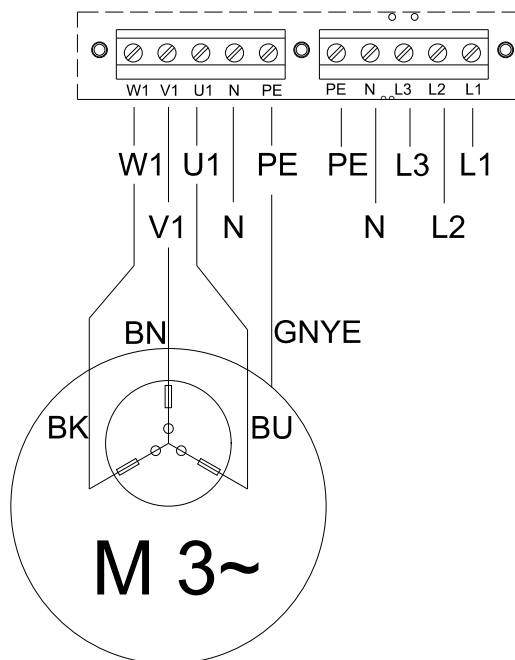


Fig. 9 Wiring diagram for three-phase pumps

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6. Startup



Warning

Before making any connections in LC 220 or working on pump, pit, etc., make sure that the power supply has been switched off and that it cannot be accidentally switched on.

Prior to startup, the connection and DIP-switch setting must have been carried out according to sections 5.3 *Electrical connection* and 5.5 *Setting*.

Startup must be carried out by authorised personnel.

Proceed as follows:

1. Check all connections.
2. Open the isolating valves in discharge and inlet lines.
3. Switch on the power supply.
4. Push the ON-OFF-AUTO selector switch to automatic mode.
5. Activate a sanitary appliance connected to the inflow of the lifting station, and monitor the increasing water level in the tank up to the start level. Check the starts and stops at least twice.

7. Maintenance and service



Warning

Before carrying out maintenance and service on lifting stations used for pumping liquids which might be hazardous to health, make sure that the lifting station has been thoroughly flushed with clean water and that the discharge pipe has been drained. Rinse the parts in water after dismantling. Make sure that the isolating valves have been closed. The work must be carried out in accordance with local regulations.



Warning

Before making any connections in LC 220 or working on lifting stations, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

Maintenance should be carried out by authorised personnel and must include electrical and mechanical maintenance.

We recommend to enter a maintenance contract with Grundfos to ensure a consistent operational safety of the system.

7.1 Electrical maintenance

- Check the gaskets of the LC 220 cabinet front cover and of the cable entries.
- Check the cable connections.
- Check the controller functions.
- Check and clean the pressure tube. See section 7.2 *Cleaning the pressure tube of the level sensor*.
- If LC 220 is installed in a particularly moist environment in a basement, we recommend checking the terminals on the PCB in order to identify possible corrosion. In typical installations, the contacts will work for many years and do not require any inspection.
- Replace the 9 V battery, if fitted, in connection with annual service.

Note

The above list is not complete. LC 220 may be installed in environments which require thorough and frequent maintenance.

7.1.1 Checking the level sensor

Check for possible leaks between pressure hose and bulkhead connector of the controller. The pressure hose must be inserted to the stop (approx. 15 mm).

The sensor's calibration is factory-set and does not need any re-calibration.

7.2 Cleaning the pressure tube of the level sensor

1. Push the ON-OFF-AUTO selector switch to position OFF (○). See section 4.2 *Control elements and indicator lights*.
2. Multilift: Loosen the screw cap by turning it counterclockwise. Unolift: The pressure tube is not fastened by means of a screw cap, but can be lifted out directly.
3. Lift the pressure tube carefully out of the collecting tank. Do not lift it by means of the hose.
4. Check for possible deposits on or in the pressure tube and the condensate trap underneath the screw cap. See section 4. *Product description*.
5. Scrape off any deposits. If necessary, remove the hose from the controller and rinse the tube and hose with clean water at low pressure.
6. Refit the pressure tube by screwing the screw cap on to the tank. Reconnect the hose to the controller.
7. Check the sensor by test running lifting station.

8. Fault finding



Warning

Before carrying out any work on lifting stations used for pumping liquids which might be hazardous to health, make sure that the lifting station has been thoroughly flushed with clean water and that the discharge pipe has been drained. Rinse the parts in water after dismantling. Make sure that the isolating valves have been closed. The work must be carried out in accordance with local regulations.

Before making any connections in LC 220 or working on lifting stations, etc., make sure that the power supply has been switched off and that it cannot be accidentally switched on.

Fault	Cause	Remedy
1. The pump does not run.	a) No power supply. Without battery backup: None of the indicator lights are on. With battery backup: See section 4. <i>Product description</i> .	Switch on the power supply.
	b) The ON-OFF-AUTO selector switch is in position OFF (○). See section 4.2 <i>Control elements and indicator lights</i> .	Push the ON-OFF-AUTO selector switch into position ON (I) or AUTO (○).
	c) Control circuit fuses are blown.	Check and eliminate the cause. Replace the control circuit fuses.
	d) The thermal switch has cut out the pump.	Allow the pump to cool down. After cooling down, the pump will restart automatically unless LC 220 has been set to manual restarting (see section 5.5 <i>Setting</i>). If so, push the ON-OFF-AUTO selector switch into position OFF (○) for a short period.
	e) Motor/supply cable is defective.	Check and replace motor and cable, if necessary.
	f) Level sensor failure.	Clean the tube of the level sensor (see section 7.2 <i>Cleaning the pressure tube of the level sensor</i>), and start up again. If the signal is still wrong, please call Grundfos service.
	g) The main board or the LED board is defective.	Replace the controller.
	h) The new DIP-switch setting does not work correctly.	Switch off the power supply to the controller for 1 minute and switch it on again (normal procedure). See section 5.5 <i>Setting</i> .
2. The pump starts and stops too frequently or even if there is no inflow.	a) Level sensor failure.	Clean the level sensor pressure tube (see section 7.2 <i>Cleaning the pressure tube of the level sensor</i>).
	b) The level sensor is blocked.	Clean the level sensor pressure tube (see section 7.2 <i>Cleaning the pressure tube of the level sensor</i>).
	c) The ventilation of the inner pump housing is blocked and the pump cannot build up pressure.	Check the pump housing and remove any impurities.
	d) Extensive foam formation in the system.	Reduce the quantity of detergent used.
3. The pump starts sometimes without visible reason.	a) Test run 24 hours after last operation.	No action necessary. It is a safety function that prevents the shaft seal from seizing up.

9. Technical data

Controller	
Voltage variants, rated voltages:	1 x 230 V, 3 x 400 V
Voltage tolerances for LC 220:	- 15 %/+ 10 % of rated voltage
Mains frequency for LC 220:	50 Hz
Supply system earthing:	For TN systems
Controller power consumption:	7 W
Backup fuse:	Depending on variant. See nameplate
Control circuit fuse:	Fine-wire fuse: 100 mA / 20 mm x Ø5
Ambient temperature:	
During operation:	0 - +40 °C (must not be exposed to direct sunlight)
In stock:	-30 - +60 °C
Enclosure class:	IP55
Cabinet of LC 220	
External dimensions:	Height = 195 mm Width = 250 mm Depth = 110 mm
Material:	ABS (acrylonitrile butadiene styrene)
Weight:	Depending on variant. See nameplate
Outputs for alarm devices:	Max. 250 VAC / max. 2 A / min. 10 mA / AC1

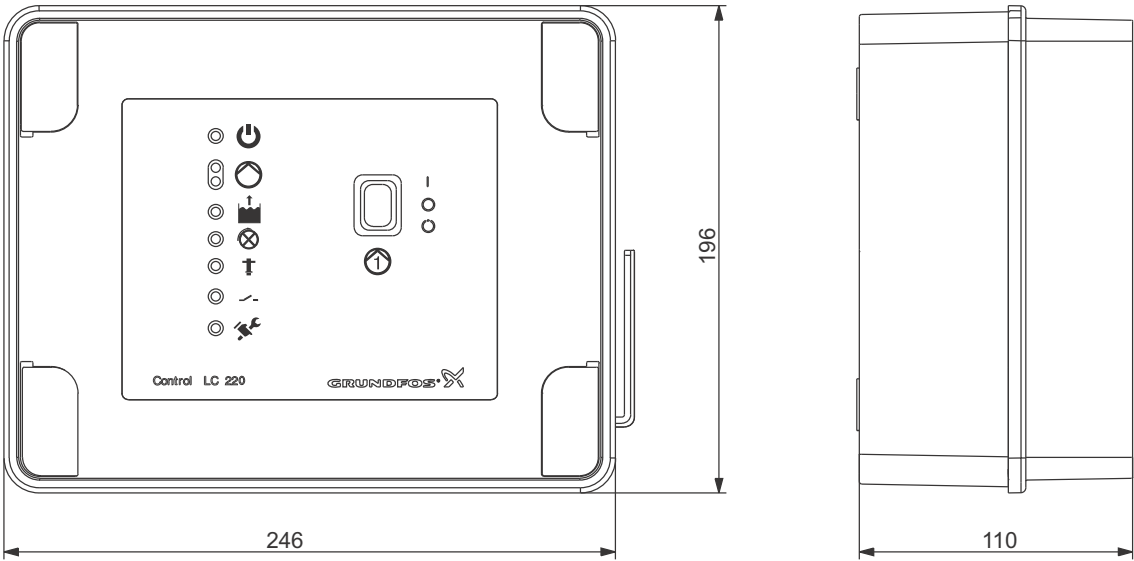
10. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

11. Dimensional drawings



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Declaration of conformity

GB: EC declaration of conformity

The LC 220 controller is part of a Multilift or Unolift system. For EC declaration of conformity, please see installation and operating instructions for the relevant system.

CZ: ES prohlášení o shodě

Řídící jednotka LC 220 je součástí systému Multilift nebo Unolift. Seznamte se s prohlášením o shodě ES uvedeným v montážním a provozním návodu příslušného systému.

DE: EG-Konformitätserklärung

Die Steuerung LC 220 ist Teil einer Multilift- oder Unolift-Anlage. Die EG-Konformitätserklärung finden Sie in der Montage- und Betriebsanleitung der entsprechenden Anlage.

GR: Δήλωση συμμόρφωσης EC

Ο ελεγκτής LC 220 αποτελεί τμήμα ενός συστήματος Multilift ή Unolift. Για τη δήλωση συμμόρφωσης ΕΚ, παρακαλούμε δείτε τις οδηγίες εγκατάστασης και λειτουργίας του σχετικού συστήματος.

FR: Déclaration de conformité CE

Le coffret de commande LC 220 fait partie d'un système Multilift ou Unolift. Pour consulter la déclaration de conformité, se reporter à la notice d'installation et de fonctionnement du système concerné.

IT: Dichiarazione di conformità CE

Il regolatore LC 220 fa parte di un sistema Multilift o Unolift. Per la dichiarazione di conformità CE, consultare le istruzioni di installazione e funzionamento del relativo sistema.

LV: EK atbilstības deklarācija

LC 220 regulators ietilpst MULTILIFT vai UNOLIFT sistēmā. Paziņojumu par atbilstību prasībām skatīt attiecīgās sistēmas uzstādīšanas un ekspluatācijas instrukcijā.

HU: EK megfeleléségi nyilatkozat

Az LC 220 vezérlő egy Unolift rendszer része. Az EU megfeleléségi nyilatkozatot lásd a vonatkozó rendszer telepítési és üzemeltetési utasításában.

UA: Декларация відповідності ЄС

Контролер LC 220 входить до складу системи Multilift або Unolift. Декларация з відповідності нормам ЄС входить до складу інструкції з монтажу та експлуатації відповідної системи.

PT: Declaração de conformidade CE

O controlador LC 220 faz parte de um sistema Multilift ou Unolift. Consulte a declaração de conformidade CE nas instruções de instalação e funcionamento do respectivo sistema.

RO: Declarație de conformitate CE

Controlerul LC 220 face parte dintr-un sistem Multilift sau Unolift. Pentru declarația de conformitate CE, consultați instrucțiunile de instalare și exploatare pentru sistemul relevant.

SI: ES izjava o skladnosti

Krmilnik LC 220 je del sistema Multilift ali Unilift. Izjava o skladnosti ES si oglejte v navodilih za montažo in obratovanje ustreznega sistema.

FI: EY-vaatimustenmukaisuusvakuutus

LC 220 -säädin on osa Multilift- tai Unolift-järjestelmää. EY:n vaatimustenmukaisuusvakuutus on luettavissa käytettävän järjestelmän asennus- ja käyttöohjeista.

TR: EC uygunluk bildirgesi

LC 220 kontrolörü, bir Multilift veya Unolift sisteminin bir parçasıdır. AT uygunluk beyanı için lütfen ilgili sistemin kurulum ve kullanım talimatlarına bakınız.

BG: ЕС декларация за съответствие

Контролерът LC 220 е част от системата Multilift или Unolift. За декларацията за съответствие на ЕС моля, вижте инструкциите за монтаж и експлоатация за съответната система.

DK: EF-overensstemmelseserklæring

LC 220-styringen er en del af enten et Multilift- eller et Unolift-anlæg. EF-overensstemmelseserklæringen fremgår af monterings- og driftsinstruktionen for det relevante anlæg.

EE: EL vastavusdeklaratsioon

LC 220 juhtplokk on osa Multilift või Unolift süsteemist. Palun vaadake EÜ vastavusdeklaratsiooni vastava seadme paigaldus- ja kasutusjuhendist.

ES: Declaración CE de conformidad

El controlador LC 220 forma parte de un sistema Multilift o Unolift. Encontrará la declaración de conformidad de la CE en las instrucciones de instalación y funcionamiento del sistema correspondiente.

HR: EZ izjava o usklađenosti

Regulator LC220 dio je sustava Multilift ili Unolift. Za EC izjavu o sukladnosti, molimo pogledajte upute za ugradnju i rad za odgovarajući sustav.

KZ: EO сәйкестік туралы мәлімдеме

LC 220 контроллері - Multilift немесе Unolift жүйесінің бөлігі. Тиісті жүйеге арналған орнату және пайдалану нұсқаулықтарынан EO сәйкестік туралы декларациясын қараңыз.

LT: EB atitikties deklaracija

LC 220 valdiklis yra "Multilift" arba "Unolift" sistemos dalis. EC atitikties deklaracija pateikta atitinkamos sistemos įrengimo ir naudojimo instrukcijoje.

NL: EC overeenkomstigheidsverklaring

De LC 220 regelaar maakt onderdeel uit van een Multilift of Unolift systeem. Voor de EG-overeenkomstigheidsverklaring raadpleegt u de installatie- en bedrijfsinstructies voor het relevante systeem.

PL: Deklaracja zgodności WE

Sterownik LC 220 stanowi część systemu Multilift lub Unolift. Patrz: deklaracja zgodności WE w instrukcji montażu i eksploatacji odpowiedniego systemu.

RU: Декларация о соответствии ЕС

Шкаф управления LC 220 является частью установок Multilift или Unolift. Декларация о соответствии ЕС включена в состав руководства по монтажу и эксплуатации насосной установки.

SK: Prehlásenie o konformite ES

Riadiaca jednotka LC 220 je súčasťou sústavy Multilift alebo Unolift. Vyhlásenie o súlade si, prosím, pozrite v montážnom a prevádzkovom návode pre príslušnú sústavu.

RS: EC deklaracija o usaglašenosti

Regulator LC 220 je deo Multilift ili Unolift sistema. Za EC deklaraciju o usklađenosti molimo pogledajte uputstvo za instalaciju i rad odgovarajućeg sistema.

SE: EG-försäkran om överensstämmelse

Styrheten LC 220 är en del av ett Multilift- eller Unolift-system. EG-försäkran om överensstämmelse finns i monterings- och driftsinstruktionen för tillämpligt system.

CN: EC 产品合格声明书

LC 220 控制器是 Multilift 或 Unolift 系统的一部分。关于 EC 合规性声明，请参阅相关系统的安装和操作说明书。

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